

Sachin Laxman Masti

Data Scientist | Machine Learning Engineer

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SUMMARY

Machine Learning Engineer who has independently shipped three production ML systems — a spam classifier, an NYC pricing model, and a churn predictor — each trained, tuned, and deployed end-to-end. Comfortable with gradient-boosted and ensemble methods (LightGBM, CatBoost, XGBoost), and deploying via FastAPI, Flask, Docker, and Gradio.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: LightGBM, CatBoost, XGBoost, Ensemble Methods, Scikit-learn, Optuna, SMOTE

Deployment & Tools: FastAPI, Flask, Docker, Docker Compose, Pydantic, Gradio, Git, GitHub, Render

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, Hypothesis Testing

PROJECTS

Spam Detection System

Text Classification · NLP

[GitHub](#) · [Live Demo](#)

- Built an end-to-end spam classifier — TF-IDF vectorization with an Optuna-tuned LightGBM classifier — on a **5,168-message** dataset (8 features).
- Corrected for class imbalance using SMOTE and validated with 5-fold cross-validation (mean ROC-AUC **99.47%**) before reaching **99.63% test AUC** and **96.8% precision** on the spam class — just **4 false positives** across 903 legitimate messages.
- Shipped as a real-time FastAPI endpoint in a Docker container, with Pydantic v2 schemas (custom field and model validators) for input sanitization.

Airbnb NYC Price Predictor

Regression · Feature Engineering

[GitHub](#) · [EDA](#) · [Live Demo](#)

- Cleaned a **48,895-listing** NYC Airbnb dataset and engineered pricing signals — Times Square proximity scoring and TF-IDF keyword detection on listing titles (“luxury,” “cozy,” “subway”).
- Tuned a CatBoost regressor to **$R^2 = 0.63$ (MAE $\approx$ \$31/listing)** on held-out data.
- Deployed on Render via Flask with a rule-based fallback estimator, keeping the app serving predictions even if the trained model fails to load.

Customer Churn Prediction System — ChurnGuard AI

Classification · Ensemble Learning

[GitHub](#) · [EDA](#) · [Live Demo](#)

- Built an end-to-end churn prediction system on a **~20,000-customer** dataset (16 attributes), with a scikit-learn pipeline handling missing-value imputation, winsorization, and one-hot/target encoding.
- Trained a bagging ensemble of 150 XGBoost classifiers, reaching **74.1% accuracy** and **75.5% precision / 75.4% recall** (F1: 75.4%) on a **4,053-customer** held-out test set — tuned the decision threshold ($p \geq 0.4658$) to balance false positives and false negatives.
- Deployed as two containerized services (FastAPI inference API + Gradio dashboard) via Docker Compose on Render, with a live insights dashboard exposing model metrics and feature distributions.

EDUCATION

Bachelor of Science, Computer Science

Shivaji University, Kolhapur · 2020–2023

WRITING

Writes technical posts on classification metrics and hyperparameter tuning — medium.com/@Sachinmasti